INTERNSHIP REPORT

Submitted

For the Partial Fulfilment of the Requirement for the Degree of Bachelor of _____ with _____ (Major and Minor Subjects) under **Magadh University**

Submitted to

The Department of _____, _____ **(College Name)**

[Paste College Logo here]

Submitted by

Name : _____

Semester : _____

Programme FYUGP : _____

University's Registration No. : _____

College Roll No. : _____

Year : _____

RK NEXORA
CAMPUS TO CAREER
A Unit of RKNexora Private Limited.

INTERNSHIP REPORT

Topic - Web Development

Submitted to the Head of Department, _____ (College Name), in partial fulfilment of the requirements for the Degree of _____, Magadh University

Name of Student : _____

Programme : _____

University Registration No. : _____

College Roll No. : _____

Internship Registration No. : _____

Name of the IPO / Agency : RKNEXORA PRIVATE LIMITED

Internship Period : _____ to _____

Semester : _____

Year / Session : _____



Signature of Student

Seal & Signature of IPO

ACKNOWLEDGEMENT

I express my sincere gratitude to RK NEXORA, RKNEXORA PRIVATE LIMITED, _____ (College Name), my faculty mentors and the internship supervisor, _____, for their guidance and support. The structured activities, learning resources and regular feedback helped me complete this internship in Web Development successfully.

I am thankful to the Head of the Department and the college mentor for their encouragement and for guiding me in preparing this report. I am also grateful to the management of RKNEXORA PRIVATE LIMITED for providing a professional learning environment and the opportunity to learn and apply my knowledge in a practical setting. The concept sessions, hands-on tasks with sample data and case studies, and the supervisor's patient and detailed feedback helped me understand how websites are planned, designed and built step by step.

Finally, I thank my family and peers for their constant encouragement throughout the programme.

_____

Name: _____

University Registration No. - _____

INTERNSHIP COMPLETION CERTIFICATE

The Internship Completion Certificate issued by RKNEXORA PRIVATE LIMITED for the internship in Web Development is to be attached on this page.

[Attach the original / printed copy of the QR-verified Internship Completion Certificate here]

Certificate No. _____

CONTENTS

Acknowledgement..... 3

Internship Completion Certificate..... 4

Preface..... 6

Executive Summary..... 7

About the Agency..... 8

Plan and Execution..... 9

Learning Outcome..... 23

Conclusion..... 25

Attendance Sheet..... 27

PREFACE

Internships are an important part of the undergraduate programme, as they allow students to see how classroom learning is applied in real organisations. The Four-Year Undergraduate Programme (FYUGP), introduced in line with the National Education Policy 2020, gives special importance to experiential learning, skill development and employability. This report has been prepared as part of the internship requirement of Magadh University and the Higher Education guidelines applicable in Bihar.

The report describes the internship undertaken in Web Development at RKNEXORA PRIVATE LIMITED, the plan followed, the work carried out, and the learning gained. Today a shop, a school, a coaching centre or a government office is judged by its website, and millions of people in Bihar first meet a business on a mobile phone screen. Web development is the skill of building these pages with HTML, CSS and JavaScript so that they look good, work on every screen and are easy to use. It is one of the most accessible technology skills: all that is needed to begin is a computer, a browser and a text editor. This report records how the basics were learnt and how a complete responsive website was designed, coded, tested and prepared for publishing. Real code and real screenshots of the pages built during the internship are included.

The report has been arranged in the order suggested by the University, beginning with the cover page and acknowledgement and ending with the conclusion and attendance sheet. It is hoped that it gives a clear and honest account of the internship experience. All examples in this report use dummy data and fictitious names; no real personal or business information has been handled.

EXECUTIVE SUMMARY

This report documents the internship completed in Web Development at RKNEXORA PRIVATE LIMITED (RK NEXORA - 'Campus to Career'). The programme required a minimum of 120 hours, completed over about four to five weeks (28 working days) with daily sessions of four to five hours, and was supported by an attendance log, supervisor guidance and review.

Background. A website is a collection of web pages stored on a server and viewed through a browser. Every page is built with three technologies: HTML gives the structure and content, CSS gives the style and layout, and JavaScript gives the behaviour. A good website is responsive, which means that it adjusts itself to the size of the screen, whether it is a large monitor or a small phone, and it is accessible, fast and easy to navigate. Because most Indian users reach the internet through mobile phones, responsive design is no longer optional.

Objectives. The internship aimed to (i) understand how the web works, (ii) write well-structured pages in HTML5, (iii) style pages with CSS3 and build layouts with Flexbox and Grid, (iv) add behaviour with JavaScript and the DOM, (v) make pages responsive with media queries, (vi) build and validate forms, (vii) learn the basics of domains and hosting, and (viii) design, build and test a small multi-page website.

Work carried out. Each topic had a concept session followed by hands-on coding in a code editor, with the result checked in a browser after every change. Pages were first built in plain HTML, then styled with CSS, then given layouts, interaction and form validation. The later part of the programme was devoted to the mini website project, a case study exercise, a mock practical exercise, supervisor review and preparation of this report.

Project. The work was tied together in a project titled **Responsive Website Development Project**. It is a responsive single-page website for a fictitious handicraft shop, with a header and navigation, a hero banner, a product grid, a contact form with JavaScript validation and a footer. It was tested on desktop and mobile widths, and its complete code and screenshots are given in this report.

Method. A blended method was followed - learn, practise, document and review - supported by a daily attendance log with day-wise remarks and regular feedback from the supervisor.

Learning gained. The student gained practical skill in HTML5 structure, CSS3 styling, Flexbox and Grid layouts, JavaScript and DOM manipulation, responsive design with media queries, form validation, browser developer tools and the basics of publishing a website, along with the habit of writing clean, indented and commented code.

Key observations. Web development is learnt by making small changes and seeing the result at once; layout problems become easy when the box model is understood; mobile-first thinking improves every page; and validation on the user's side saves time but must never be the only protection of data.

Conclusion. The internship met its objectives and contributed to the student's academic understanding, employability and confidence. The report follows the University's suggestive format: cover page, acknowledgement, certificate, contents, preface, executive summary, about the agency, plan and execution, learning outcome, conclusion and attendance sheet.

ABOUT THE AGENCY

RK NEXORA, a unit of RKNEXORA PRIVATE LIMITED, provides structured internship and skill-development programmes under the theme 'Campus to Career'. The organization focuses on academic-industry alignment, practical exposure, assessment, documentation and employability-oriented learning.

Particulars	Details
Name of Agency	RKNEXORA PRIVATE LIMITED
Programme / Unit	RK NEXORA - Campus to Career
Registered Office	C/O Mukesh Kumar, Khiri, PS Atri, Jethiyan, Atri, Gaya - 805236, Bihar, India
CIN	U85499BR2026PTC086757
Email	rknexorapvtltd@gmail.com
Website	www.rknexora.org
Internship Topic	Web Development

1. Vision

To support students in developing practical, professional and industry-relevant capabilities.

2. Mission

To deliver structured learning experiences that connect academic knowledge with workplace expectations.

3. Activities of the Agency

1. Structured internship programmes aligned with university and higher-education requirements.
2. Skill-development learning modules, practical activities and assignments.
3. Attendance tracking, supervision, assessment and documentation.
4. Employability-oriented guidance connecting campus learning to career readiness.

4. Programme Structure

The internship programme is organised around a fixed number of learning hours, completed over a defined period. Each intern is registered on the internship portal with a unique registration number, is assigned a supervisor, follows a day-wise plan and maintains an attendance log. Learning is organised into three phases: **Orientation** (Days 1-2), **Learning and Practice** (Days 3-22, ten sub-topics of two days each: one concept day and one hands-on day) and **Project and Review** (Days 23-28), followed by a concluding review.

5. Role of the Supervisor

The internship supervisor assigned by the agency guided the student through the programme, explained concepts, assigned practice tasks, reviewed the daily log and gave feedback. The supervisor also verified attendance and reviewed the student's work at the end of the programme.

PLAN AND EXECUTION

1. Objectives

The internship was planned around the following objectives, each linked to a part of the programme:

1. Understand how the web works - browsers, servers, URLs and HTTP (Week 1).
2. Learn HTML structure - headings, lists, links, images, tables and forms (Week 1).
3. Learn CSS styling - selectors, colours, fonts, the box model, Flexbox and Grid (Weeks 1-2).
4. Learn JavaScript basics and DOM manipulation - variables, conditions, loops, functions, events and updating the page (Week 3).
5. Learn responsive design and form validation (Week 3-4).
6. Learn hosting basics - domains, hosting options and publishing (Week 4).
7. Design, build, test and document a responsive mini website (Weeks 4-5).

2. Methodology

The programme followed a blended method made up of four connected steps, repeated for each topic:

Learn. Each topic began with a concept session in which the supervisor explained the idea, its main terms and how it is used in real organisations. Notes were prepared in the student's own words.

Practise. The concepts were then applied in guided hands-on tasks - writing the structure of a page with semantic tags, styling it with an external style sheet, building a navigation bar with Flexbox and a product grid with CSS Grid, changing page content with JavaScript events, writing media queries for phone and desktop widths, validating a contact form and checking the page in the browser's developer tools. Only sample data and fictitious examples were used, so that no real personal or business information was handled.

Document. Every day's work was recorded in the internship attendance log with a short remark, and sheets, files and screenshots were saved so that the work could be reviewed.

Review. The supervisor reviewed the work, pointed out errors and suggested improvements. Review days, a case study and a mock practical exercise helped to consolidate the learning.

3. Understanding Web Development - Concepts Studied

3.1 How the web works

When a person types a web address (URL) such as www.example.com in the browser, the browser asks a DNS (Domain Name System) server for the numeric address of the website's server. It then sends an HTTP request to that server, and the server sends back a response containing HTML, CSS, JavaScript and images, which the browser combines and displays. HTTP is the set of rules for this conversation, and HTTPS is its secure form in which the data is encrypted. A request may be GET (to fetch a page) or POST (to send data such as a form). The status code of the response tells the result: 200 means success, 404 means page not found and 500 means a server error.



Figure 1: What happens when a web page is opened

3.2 The three building blocks: HTML, CSS and JavaScript

HTML (HyperText Markup Language) describes the content of a page through tags such as <h1> for a heading, <p> for a paragraph, <a> for a link, <img> for an image and <form> for a form. HTML5 added semantic tags such as <header>, <nav>, <section>, <article> and <footer>, which describe the meaning of each part and help search engines and screen readers. CSS (Cascading Style Sheets) controls colours, fonts, spacing and layout; a rule has a selector, which chooses the elements, and declarations, which set properties. JavaScript is a programming language that runs in the browser and makes the page interactive by responding to events such as clicks and typing.

A useful way to remember the three is to think of a house: HTML is the walls and rooms, CSS is the paint and furniture, and JavaScript is the electricity that makes things work. The three are kept in separate files - index.html, style.css and script.js - so that each can be changed without disturbing the others.

3.3 The box model, Flexbox and Grid

Every element on a page is a rectangular box made of content, padding, border and margin. Understanding this box model is the key to spacing and alignment; the property box-sizing: border-box makes widths easier to calculate. Flexbox arranges items in one direction, in a row or a column, and is ideal for navigation bars and centring. CSS Grid arranges items in rows and columns together and is ideal for product listings and page layouts. A responsive design uses flexible units (percentages, fr, rem) and media queries, which apply different CSS rules when the screen width is below or above a limit.

3.4 Responsive design and accessibility

Because people use phones, tablets and computers, a page must adapt. The viewport meta tag tells a phone browser to use the real screen width, and media queries such as @media (max-width: 700px) change the layout for small screens - for example, turning a three-column grid into a single column. Accessibility means that everyone, including people who use screen readers or cannot use a mouse, can use the site. Good practice includes meaningful headings, alt text on images, labels for form fields, sufficient colour contrast and a logical tab order.

3.5 Common HTML tags and the browser developer tools

The tags used most often during the internship are summarised in the table below. The browser's developer tools, opened with the F12 key, were used daily: the Elements tab shows the page's HTML and the CSS applied to each element and allows changes to be tried live; the Console tab shows JavaScript errors and allows code to be tested; the Network tab shows each file loaded, its size and its status code; and the device toolbar shows the page at phone and tablet widths.

Tag	Purpose	Example
<h1> to <h6>	Headings of different levels	Our Products
	Paragraph of text	Hand-painted folk art.
<a>	Link to another page or section	Contact
<img>	Image with alt text	<img src="lamp.jpg" alt="Bamboo lamp">
, ,	Unordered and ordered lists	Madhubani Painting
<table>	Data in rows and columns	<tr><td>Rs. 850</td></tr>
<form>, <input>, <label>	Collecting user input	<input type="email" id="email">
<header>, <nav>, <section>, <footer>	Semantic page areas	<nav> ... </nav>

4. Week-wise Plan and Execution

Week	Days	Activities
Week 1	Days 1-6	Orientation and foundations: company induction, how the web works, HTML structure and CSS styling, each with a study session and a practice task.
Week 2	Days 7-12	Layout and interaction: CSS Flexbox and Grid, JavaScript basics and DOM manipulation.
Week 3	Days 13-18	Responsive design with media queries, forms and validation, and hosting basics.
Week 4	Days 19-24	Responsive website project: planning, coding, testing on desktop and mobile; case study exercise and mock practical exercise.
Week 5	Days 25-28	Project closure: supervisor review and feedback, internship report drafting, presentation preparation and final review.

5. Week-wise Narrative

Week 1: The first week was devoted to orientation and foundations. After company induction, the student studied how the web works and wrote the first pages in HTML using headings, lists, links, images, tables and forms. CSS was then introduced, and the same pages were styled with colours, fonts and spacing.

Week 2: In the second week the student learnt Flexbox and CSS Grid and used them to build a navigation bar and a card grid. The week then moved to JavaScript - variables, conditions, loops, functions and events - and to DOM manipulation, such as changing text and styles when a button is clicked.

Week 3: The third week was devoted to responsive design, where media queries were used to adapt layouts to phones; to forms and validation using JavaScript and regular expressions; and to hosting basics - domain names, hosting types and the steps to publish a site.

Week 4: In the fourth week the student planned and built the responsive website, tested it in the browser at desktop and mobile widths, corrected layout and validation errors, and completed a case study exercise and a mock practical exercise.

Week 5: The last week was devoted to project closure: supervisor review and feedback, drafting of the internship report, preparation of a presentation and final review before submission.

6. Topic-wise Learning in Detail

6.1 Web Fundamentals

Concept: The structure of the web, browsers, servers, URLs, IP addresses, DNS and the HTTP request-response cycle. **Practice carried out:** Using the browser's developer tools to inspect a page's HTML, view the network requests and the status codes, and drawing the request-response flow.

6.2 HTML Structure

Concept: Tags, headings, lists, links, images, tables and forms, and semantic HTML5 elements. **Practice carried out:** Building a profile page and a product page with header, nav, section and footer, an ordered and an unordered list, a table of prices and a form.

6.3 CSS Styling

Concept: Selectors (element, class, id), colours, fonts, the box model, backgrounds and hover effects. **Practice carried out:** Linking an external style sheet and styling the pages with a colour scheme, readable fonts, spacing and button hover effects.

6.4 Flexbox and Grid

Concept: Flexbox for rows and columns and CSS Grid for two-dimensional layouts, with gap, alignment and fractions. **Practice carried out:** Creating a navigation bar with display: flex and a three-column product grid with grid-template-columns: repeat(3, 1fr).

6.5 JavaScript Basics

Concept: Variables (let, const), data types, conditions, loops, functions and events. **Practice carried out:** Writing a tip calculator, an even-odd checker and a function that finds the largest number in an array.

6.6 DOM Manipulation

Concept: The Document Object Model, selecting elements with getElementById and querySelector and changing text, style and classes. **Practice carried out:** Making a button that changes the page's theme, a counter and a to-do list that adds and removes items.

6.7 Responsive Design

Concept: Viewport meta tag, flexible units, media queries and mobile-first thinking. **Practice carried out:** Changing the three-column grid to one column below 700 pixels and testing at several widths in the developer tools.

6.8 Forms and Validation

Concept: Input types, labels, placeholders and validating name, email and mobile number with JavaScript and regular expressions. **Practice carried out:** Building the contact form with error messages shown below each field (see section 10).

6.9 Hosting Basics

Concept: Domain names, shared, cloud and static hosting, FTP and static site publishing, HTTPS and SEO basics. **Practice carried out:** Preparing the project folder for publishing, checking all links and image paths, and writing a short publishing checklist.

6.10 Mini Website Project

Concept: Planning, coding, testing and documenting a small multi-page website. **Practice carried out:** Building the Gaya Craft Corner website described in section 10.

7. Good Practice, Standards and Ethics - Points Studied

A good website follows standards and respects its users. The points studied are summarised below.

- **Semantic and valid HTML:** use headings in order and correct tags; check the code with a validator.
- **Separation of concerns:** keep structure (HTML), style (CSS) and behaviour (JavaScript) in separate files.
- **Accessibility:** alt text for images, labels for form fields, good colour contrast and keyboard-friendly pages.
- **Performance:** small, compressed images and minimal code make pages load quickly on slow mobile networks.
- **Security:** never trust data typed into a form - validation in the browser is only for convenience, and the server must check again; use HTTPS.
- **Privacy and consent:** collect only the data that is needed and explain why; the Digital Personal Data Protection Act, 2023 applies to personal data.
- **Copyright and licences:** use only images, fonts and code that are free to use or properly licensed, and acknowledge the source.

8. Process Flows Studied

8.1 Website development cycle

- **Plan:** decide the purpose, the audience and the pages; draw a simple wireframe on paper.
- **Design:** choose colours, fonts and the layout for mobile and desktop.
- **Build:** write the HTML structure, then the CSS, then the JavaScript.
- **Test:** check in different browsers and screen widths; test every link and form.
- **Publish:** upload the files to a host and connect the domain name.
- **Maintain:** update content, fix problems and keep a backup.

8.2 Debugging a page that looks wrong

- Open the developer tools (F12) and use the Elements tab to see which CSS rule is applied.
- Check the Console tab for JavaScript errors and the line number.
- Check for spelling mistakes in class names, missing closing tags and missing semicolons or brackets.
- Change one thing at a time and refresh the page.
- Test at a narrow width to find responsive problems.

9. Worked Examples with Sample (Dummy) Data

Worked example - a tip calculator in JavaScript. To practise the DOM, a small calculator was written. The function reads the bill and the tip percentage from the page, calculates the tip and the total, and writes the result back into the page. The same logic is shown below as a plain function, with the values obtained when it was tested in the browser console.

```
function calcTip(bill, percent) {
  const tip = bill * percent / 100;
  return { tip: tip, total: bill + tip };
}
console.log(calcTip(850, 10));   // { tip: 85, total: 935 }
console.log(calcTip(450, 5));    // { tip: 22.5, total: 472.5 }
```

The results were checked by hand: 10% of 850 is 85, so the total is 935, and 5% of 450 is 22.5, so the total is 472.5. In the page version, document.getElementById('tipResult').textContent = ... updated the screen without reloading it, which is the main idea of DOM manipulation. Another point learnt in this exercise was CSS specificity: when two rules apply to the same element, an id selector beats a class selector, which beats an element selector, and a later rule beats an earlier one of equal strength. Understanding this removed many 'my style is not working' problems.

Worked example - CSS units. The project is described in full in section 10. In this section a simpler exercise is shown: how the three kinds of CSS length units behave when the screen changes.

Unit	Type	Meaning	Typical use
px	Fixed	One screen pixel	Borders, small details
%	Relative	Percentage of the parent element	Widths of columns
rem	Relative	Multiple of the root font size	Font sizes and spacing
fr	Relative	Fraction of free space in a grid	Grid columns
vw / vh	Relative	1% of viewport width / height	Full-screen sections

The exercise showed that a layout built with relative units adjusts itself automatically, whereas a layout built only with pixels breaks on small screens. The following table summarises common mistakes met while building pages and how they were fixed.

Problem	Reason	Solution
Style not applied	Wrong file path in the link tag or misspelt class name	Correct the path and the name; check in developer tools
Elements overflow the screen	Fixed width in pixels	Use max-width: 100% or flexible units
Image too large	No size limit set	Set width: 100% and height: auto
Page too small on phone	Viewport meta tag missing	Add the viewport meta tag
Button does nothing	Script file not linked or wrong id	Link script.js before </body>; check ids
Form refreshes the page	Default submit action	Use event.preventDefault() in the handler

10. Project: Responsive Website Development Project

Problem statement. A small handicraft shop in Gaya, selling Madhubani paintings, bamboo lamps and sikki grass baskets, wants a simple website that looks good on phones and computers, shows its products and lets customers send a message. The shop is fictitious and the products and prices are dummy.

10.1 Requirements and design

- A header with the shop name and a navigation bar linking to the About, Products and Contact sections.
- A hero banner with a short message and a button.
- A product section with three cards in a grid (one column on mobile).
- A contact form with name, email and mobile number, validated with JavaScript.
- A footer; the page must work on both desktop and phone widths.

The page was planned as a single page with three sections and designed with a dark navy header, an orange-red gradient hero and white cards with a brown price colour. The structure was written first in HTML, then styled in CSS, and finally given behaviour in JavaScript.

10.2 The code - index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Gaya Craft Corner</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <header class="site-header">
    <h1 class="logo">Gaya Craft Corner</h1>
    <nav>
      <a href="#about">About</a>
      <a href="#products">Products</a>
      <a href="#contact">Contact</a>
    </nav>
  </header>

  <section class="hero">
    <h2>Handmade crafts from Bihar</h2>
    <p>Madhubani paintings, bamboo ware and sikki grass baskets.</p>
    <a class="btn" href="#products">View Products</a>
  </section>

  <section id="about" class="container">
    <h2>About Us</h2>
    <p>We are a small demo shop that shows how a responsive website is built with
HTML5, CSS3 and JavaScript.</p>
  </section>

  <section id="products" class="container">
    <h2>Our Products</h2>
    <div class="grid">
      <article class="card"><h3>Madhubani Painting</h3><p>Hand-painted folk
art.</p><span class="price">Rs. 850</span></article>
      <article class="card"><h3>Bamboo Lamp</h3><p>Eco-friendly table lamp.</p><span
class="price">Rs. 450</span></article>
      <article class="card"><h3>Sikki Basket</h3><p>Woven grass basket.</p><span
class="price">Rs. 320</span></article>
    </div>
  </section>

  <section id="contact" class="container">
    <h2>Contact Us</h2>
    <form id="contactForm" novalidate>
      <label for="name">Name</label>
```

```
      <input type="text" id="name" placeholder="Your name">
      <small class="error" id="nameError"></small>

      <label for="email">Email</label>
      <input type="email" id="email" placeholder="you@example.com">
      <small class="error" id="emailError"></small>

      <label for="phone">Mobile number</label>
      <input type="text" id="phone" placeholder="10-digit number">
      <small class="error" id="phoneError"></small>

      <button type="submit" class="btn">Send Message</button>
      <p id="formMessage"></p>
    </form>
  </section>

  <footer>&copy; 2026 Gaya Craft Corner (demo website)</footer>
  <script src="script.js"></script>
</body>
</html>
```

10.3 The code - style.css

The style sheet uses a universal reset, Flexbox for the header, CSS Grid for the product cards and a media query for screens up to 700 pixels wide.

```
* { box-sizing: border-box; margin: 0; padding: 0; }
body { font-family: Arial, sans-serif; color: #1f2937; line-height: 1.6; }

.site-header {
  display: flex; justify-content: space-between; align-items: center;
  padding: 14px 6%; background: #1f2937; color: #fff;
}
.site-header nav a { color: #fff; text-decoration: none; margin-left: 20px; }
.site-header nav a:hover { color: #fbbf24; }

.hero {
  text-align: center; padding: 70px 6%;
  background: linear-gradient(135deg, #f59e0b, #dc2626); color: #fff;
}
.hero p { margin: 10px 0 20px; }

.btn {
  display: inline-block; padding: 10px 22px; border: none; cursor: pointer;
  background: #1f2937; color: #fff; border-radius: 6px; text-decoration: none; font-
size: 1rem;
}

.container { padding: 40px 6%; }
.container h2 { margin-bottom: 14px; }

.grid { display: grid; grid-template-columns: repeat(3, 1fr); gap: 20px; }
.card { border: 1px solid #d1d5db; border-radius: 8px; padding: 18px; }
.card .price { display: block; margin-top: 8px; font-weight: bold; color: #b45309; }

form { max-width: 460px; display: flex; flex-direction: column; }
label { margin-top: 12px; font-weight: bold; }
input { padding: 9px; border: 1px solid #9ca3af; border-radius: 5px; font-size:
1rem; }
input.invalid { border-color: #dc2626; }
.error { color: #dc2626; min-height: 18px; }
#formMessage { margin-top: 10px; color: #15803d; font-weight: bold; }
form .btn { margin-top: 16px; align-self: flex-start; }

footer { text-align: center; padding: 18px; background: #f3f4f6; font-size: 0.9rem; }

@media (max-width: 700px) {
  .site-header { flex-direction: column; gap: 8px; }
```

```
  .grid { grid-template-columns: 1fr; }
}
```

10.4 The code - script.js

The script reads the form fields when the form is submitted, validates each one (name of at least three letters, a valid email pattern and a 10-digit Indian mobile number starting with 6 to 9), shows an error below the wrong fields and shows a thank-you message when everything is correct.

```
const form = document.getElementById("contactForm");

function showError(id, message) {
  document.getElementById(id + "Error").textContent = message;
  document.getElementById(id).classList.toggle("invalid", message !== "");
}

form.addEventListener("submit", function (event) {
  event.preventDefault();
  const name = document.getElementById("name").value.trim();
  const email = document.getElementById("email").value.trim();
  const phone = document.getElementById("phone").value.trim();
  let ok = true;

  showError("name", name.length < 3 ? "Please enter at least 3 letters." : "");
  showError("email", /^[^\s@]+@[^\s@]+\.[^\s@]+$/.test(email) ? "" : "Enter a valid
email address.");
  showError("phone", /^[6-9][0-9]{9}$/.test(phone) ? "" : "Enter a valid 10-digit
mobile number.");

  ok = name.length >= 3 && /^[^\s@]+@[^\s@]+\.[^\s@]+$/.test(email) && /^[6-9][0-9]
{9}$/.test(phone);
  document.getElementById("formMessage").textContent =
    ok ? "Thank you, " + name + "! Your message has been sent." : "";
});
```

10.5 Output - desktop and mobile views

The page was opened in the browser and tested at a desktop width of 1100 pixels and a phone width of 390 pixels. On the desktop the three cards appear in one row; on the phone the navigation moves below the logo and the cards stack in one column, as the media query intends.

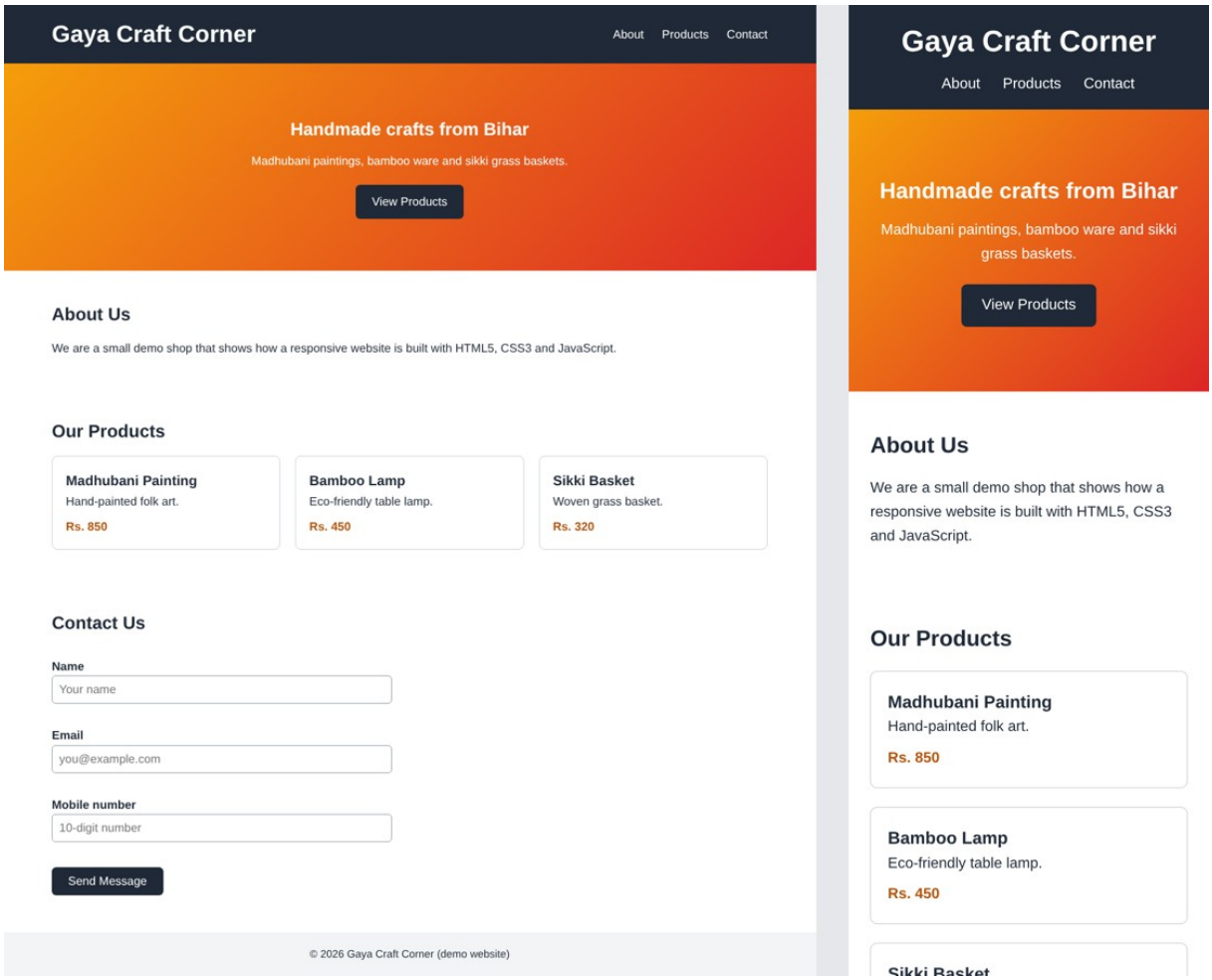


Figure 2: The Gaya Craft Corner website - desktop view (left) and mobile view (right)

10.6 Output - form validation

The form was tested with wrong values (a name of two letters, the email 'ravi@com' and the number '12345'). The page showed an error message below each field and a red border, as in the first figure. With correct values the thank-you message appeared.

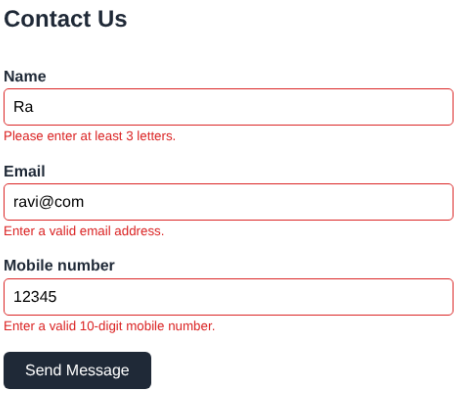


Figure 3: Validation errors for wrong input

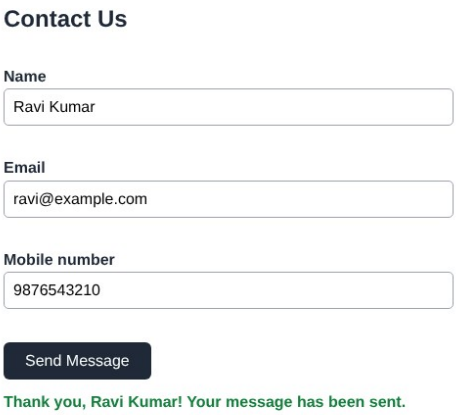


Figure 4: Success message for correct input

10.7 Testing and results

The page was tested on different screen widths and browsers. The table below records the results of the test, and a few extra checks on speed and accessibility are listed after it.

Test	Width / browser	Expected result	Result
Desktop layout	1100 px, Chrome	Three product cards in a row	Passed
Tablet layout	768 px, Chrome	Cards in a row, nav in one line	Passed
Phone layout	390 px, Chrome	Nav below logo, one card per row	Passed
Form - wrong data	Chrome and Firefox	Three error messages in red	Passed
Form - correct data	Chrome and Firefox	Thank-you message shown	Passed
Links	All sizes	Jump to the correct section	Passed

- Page size is under 10 KB because no images are used, so it loads instantly even on slow mobile data.
- The form fields have labels, and the colour contrast of text on the dark header and white cards is good.
- Navigation links jump to the correct sections on desktop and phone.
- The grid changes from three columns to one at 700 pixels.
- All three fields show errors for wrong input, and no message is shown if all are correct.
- The page works in Chrome and Firefox; images are not used, so it loads very quickly.
- Possible improvements: product images, a mobile menu button, saving the form data on a server, and a separate page for each product.

11. Case Study Exercise: A Local Shop Without a Website

A fictitious sweet and snacks shop in a Bihar town has a loyal group of local customers but gets almost no new customers. Its owner says that people ask for the menu and prices on the phone all day, that competitors appear first when people search online, and that the shop's only 'website' is a social media page that is not updated. The student was asked to suggest a web solution.

Analysis. The shop needs a clear online presence: a menu with prices, the address and timings, a map, a phone and WhatsApp button, and photographs. Most of its visitors will use a mobile phone on a slow connection, so the site must be light and responsive.

Solution proposed. A simple three-page responsive website - Home, Menu and Contact - built with HTML, CSS and a little JavaScript, using compressed images, a click-to-call button and an embedded map; published on low-cost static hosting with a domain name and HTTPS; and registered on the Google business listing. A form for festival-order enquiries would be added later.

Expected benefit and cautions. Phone enquiries would be reduced, new customers would find the shop, and festival orders could be collected in advance. The cautions noted were to keep the menu and prices updated, to get customers' permission before using their photographs and to protect any phone numbers collected.

12. Mock Practical Exercise: Building and Testing a Small Website in Fixed Time

A mock practical exercise was conducted at the end of the learning phase to check that the student could build a small website on her own within a fixed time. The task was to build a one-page 'Class Notice Board' site with a header, three notice cards and a feedback form. The steps followed were:

1. Planning: the sections were listed and a quick wireframe was drawn for mobile and desktop.
2. HTML: semantic tags were used for header, main, section, article and footer, with headings in the right order.
3. CSS: a colour scheme, fonts, spacing and a Flexbox navigation bar were applied; the cards were placed in a grid.
4. Responsive design: a media query was added to stack the cards on small screens.
5. JavaScript: a button to hide and show a notice and a feedback form with validation were added.
6. Testing: the page was opened at three widths and every link and field was tested.
7. Review: the supervisor checked the structure, naming, indentation and accessibility, and suggested adding alt text and labels, which were done.
8. Presentation: the page and its code were explained to the supervisor in about five minutes.

13. Sample Checklists Prepared

Page quality checklist

- Every page has a title, a language attribute and the viewport meta tag.
- Headings are in order and images have alt text.
- Links and buttons work; the code is indented and commented.
- Colour contrast is good and text is readable on a phone.

Testing and publishing checklist

- Tested at desktop, tablet and phone widths.
- Form validation tested with correct, wrong and empty values.
- No errors in the browser console; file names and paths are correct (case-sensitive).
- Backup taken before publishing; HTTPS enabled.

14. Tools and Resources

The following tools and resources were used during the internship: a code editor (Visual Studio Code), the Google Chrome browser with its developer tools, an HTML and CSS validator, Playwright for taking screenshots of pages, MS Word for the report, the MDN Web Docs and W3C documentation, learning material and practice tasks given by the agency, and regular mentor feedback.

15. Programme at a Glance

The programme was in the domain of Web Development with RKNEXORA PRIVATE LIMITED as the agency. It required a minimum of 120 hours over 28 learning days, organised in three phases: Orientation (Days 1-2), Learning and Practice (Days 3-22) and Project and Review (Days 23-28). Attendance was recorded in the Internship Attendance Log, signed by the supervisor (see Attendance Sheet).

The approximate distribution of the effort across the topics is shown below.

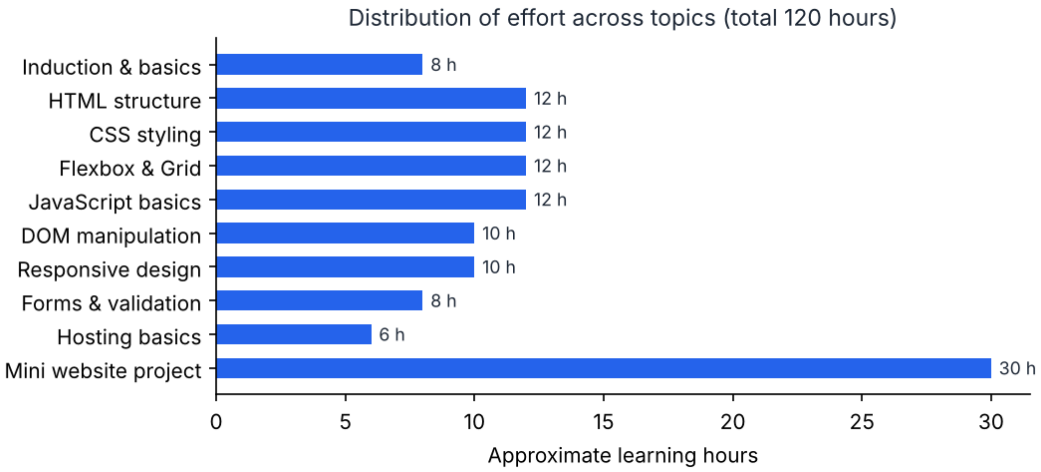


Figure 5: Distribution of learning effort across topics (120 hours)

LEARNING OUTCOME

a. Knowledge Acquired

1. How the web works: browsers, servers, URLs, DNS, HTTP and HTTPS, request methods and status codes.
2. HTML5 structure: headings, paragraphs, lists, links, images, tables, forms and semantic elements.
3. CSS3 styling: selectors, colours, fonts, the box model, backgrounds, hover effects and units.
4. Layouts with Flexbox and CSS Grid.
5. JavaScript basics: variables, data types, conditions, loops, functions and events.
6. DOM manipulation: selecting elements and changing text, styles and classes.
7. Responsive design with the viewport tag and media queries.
8. Form elements and validation with regular expressions.
9. Domain names, types of hosting and the steps of publishing a site; basic SEO and accessibility ideas.
10. Planning, building, testing and documenting a complete small website.

Glossary of Terms Learned

- **HTML / CSS / JavaScript** - Structure / style / behaviour of a web page.
- **Tag / Attribute** - A markup instruction such as <p> / extra information inside a tag, such as href.
- **Semantic HTML** - Tags that describe meaning, such as header and nav.
- **Selector** - The part of a CSS rule that chooses elements.
- **Box model** - Content, padding, border and margin of an element.
- **Flexbox / Grid** - One-dimensional / two-dimensional CSS layout systems.
- **Responsive design** - A design that adjusts to the screen size.
- **Media query** - A CSS rule that applies only at certain screen sizes.
- **Viewport** - The visible area of a page on a device.
- **DOM** - The tree of page elements that JavaScript can read and change.
- **Event** - An action such as a click or a key press.
- **Regular expression** - A pattern used to check text such as an email.
- **URL / DNS / HTTP** - Web address / address book of the internet / rules of web communication.
- **Domain name** - The name of a website, such as example.com.
- **Hosting** - A server space where a website's files are stored.
- **SEO** - Search Engine Optimisation, improving visibility in search results.
- **Alt text** - A text description of an image for screen readers.
- **Validation** - Checking that input is correct before accepting it.

b. Skills Learned

Technical skills. Writing structured HTML5; styling with CSS3 and building layouts with Flexbox and Grid; writing JavaScript functions and event handlers; manipulating the DOM; making pages responsive; validating forms with regular expressions; using the browser's developer tools; testing on different screen sizes; and preparing a site for publishing.

Professional skills. Attention to detail; logical and visual thinking; problem solving and debugging; time management; clear documentation; and the ability to explain a design and code to others.

These skills were developed through repeated practice and daily documentation. Rebuilding the same page several times improved accuracy, using the developer tools built debugging skill, testing on a phone width built user-focused thinking, and explaining the project to the supervisor built confidence.

c. Observed Attitude and Gained Values

1. **Patience and attention to detail:** a missing bracket or quote can break a whole page, so careful checking became a habit.
2. **User focus:** every design decision was checked from the user's side - is it readable, fast and easy on a phone?
3. **Honesty:** code and images were used with proper permission and acknowledgement.
4. **Respect for privacy:** only dummy data was used in forms.
5. **Discipline and punctuality:** regular reporting at fixed timings and daily attendance records.

d. Challenges Faced

Understanding the box model and why elements did not line up was handled by using the developer tools to see margins and padding, and by setting box-sizing: border-box.

Choosing between Flexbox and Grid was handled by using Flexbox for one-direction layouts and Grid for two-direction layouts, with small practice examples of each.

Making the layout work on a phone was handled by testing at narrow widths from the start and using a media query to stack the cards.

Understanding JavaScript events and the DOM was handled by practising small examples such as a counter and a theme button.

Writing the regular expressions for email and mobile numbers was handled by testing them with many sample inputs.

Managing time between coding, documentation and report writing was handled by dividing work week by week and using review days for corrections.

e. Career Pathways

The knowledge gained supports careers such as Front-end Web Developer (junior), Web Designer, WordPress and CMS Developer, UI Developer, Website Content and SEO Executive, Freelance Web Developer, and, with further study, Full Stack Developer. Opportunities exist in IT companies, digital agencies, start-ups, e-commerce, media and government digital projects, and many small businesses in Bihar need simple websites. Related higher-study options include BCA, MCA, B.Tech in computer science and information technology and certificate courses in front-end and full stack development. A good next step is to build three or four small sites, publish them online, learn Git and a framework such as React, and keep the projects as a portfolio.

CONCLUSION

The internship in Web Development was a valuable learning experience that integrated academic knowledge with practical activity. The programme achieved its objectives through structured learning, guided practice, assignments, supervisor review and documentation.

The student gained a clear understanding of how websites are built - through HTML structure, CSS styling and layouts, JavaScript behaviour, responsive design, form validation and hosting - and acquired practical experience by building and testing a complete responsive website. The experience has strengthened the student's employability and provides a foundation for further study in front-end and full stack development.

Reflection

Looking back, the most useful lesson of the internship was that web pages are built in small steps and improved by constant testing. Seeing a change appear in the browser at once made learning enjoyable, and checking the page on a phone width taught that design is for real users. Mistakes such as a missing closing tag or a wrong file path taught the habit of careful, patient checking, and recording each day's work made the progress visible over the weeks.

Objectives and Achievement

The objective of understanding the web and writing HTML was achieved through the profile and product pages. The objective of CSS and layouts was achieved through the styled pages, the Flexbox navigation and the Grid cards. The objective of JavaScript and the DOM was achieved through the calculator, counter and theme button. The objective of responsive design and forms was achieved through the media query and the validated contact form. The objective of building a complete site was achieved through the Gaya Craft Corner project, tested on desktop and mobile widths.

Future Scope

The knowledge gained can support further study of front-end and full stack development, UI design and digital marketing, or freelance work building websites for small businesses. With the growth of mobile internet, e-commerce and digital government services in Bihar and across India, people who can build fast, accessible and responsive websites are increasingly needed.

Suggestions for Improvement of the Programme

1. Include a session on version control with Git and publishing on a free static host.
2. Add a short module on accessibility testing and basic SEO.
3. Provide a small group project in which a team builds a multi-page site.
4. Continue regular mentor feedback and progress reviews.
5. Encourage students to maintain detailed daily documentation.

References

1. Kolb, D. A. (1984). Experiential Learning: Experience as the Source of Learning and Development. Prentice Hall.
2. University Grants Commission (UGC). Internship guidelines for higher education institutions.
3. Ministry of Education, Government of India. National Education Policy 2020.
4. Mozilla Developer Network (MDN). MDN Web Docs - HTML, CSS and JavaScript. developer.mozilla.org.
5. World Wide Web Consortium (W3C). Web Content Accessibility Guidelines (WCAG). w3.org.

6. Duckett, J. HTML and CSS: Design and Build Websites. Wiley.
7. Flanagan, D. JavaScript: The Definitive Guide. O'Reilly Media.
8. Marcotte, E. Responsive Web Design. A Book Apart.
9. Government of India. The Digital Personal Data Protection Act, 2023.
10. RK NEXORA internship programme handbook.

ATTENDANCE SHEET

The Internship Attendance Log issued by the IPO, with the supervisor's signature for each day, is to be attached on this page.

[Attach the signed Internship Attendance Log here]